

CE 310 — Week 13 Pre-Lab Quiz

Week: 13 | Topic: Monte Carlo Simulation Points: 10 questions $\times$ 1 pt each = 10 pts
Submission: Complete in D2L before class begins. No partial credit for MC.

Instructions

Select the single best answer for each question. The quiz covers the purpose of Monte Carlo simulation, random sampling, analytical vs. simulated uncertainty, percentile interpretation, and variance decomposition. Review the Week 13 Background reading before class.

Questions

Q1. Monte Carlo simulation differs from one-at-a-time (OAT) sensitivity analysis primarily because Monte Carlo:

- (A) Requires fitting a new regression model for each scenario
 - (B) Varies all uncertain inputs simultaneously according to probability distributions, capturing their combined effect on output uncertainty
 - (C) Replaces the regression model with a lookup table of pre-computed values
 - (D) Only works when all inputs are normally distributed
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Q2. When a Python script begins with `np.random.seed(42)`, this means:

- (A) Only 42 random numbers will be generated during the simulation
 - (B) The random number sequence is fixed, so every run of the script produces identical results — enabling reproducibility
 - (C) All random draws will be positive integers near 42
 - (D) The simulation will converge after exactly 42 iterations
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Q3. The annual cooling model is: $\text{annual_kWh} = 166,107 + \text{delta_T} \times 8,760 \times 1.026$

where $\text{delta_T} \sim \text{Normal}(0, 2^\circ\text{F})$. What is $E[\text{annual_kWh}]$, the expected value of annual cooling?

- (A) $166,107 + 2 \times 8,760 \times 1.026 = 184,081$ kWh
- (B) 166,107 kWh — because $E[\text{delta_T}] = 0$, so the delta term contributes zero in expectation
- (C) $2 \times 8,760 \times 1.026 = 17,975$ kWh (only the delta term counts)
- (D) The expected value cannot be determined without running the simulation

Q4. Using the same model $\text{annual_kWh} = 166,107 + \text{delta_T} \times 8,760 \times 1.026$ with $\text{delta_T} \sim \text{Normal}(0, 2^\circ\text{F})$, what is the analytical standard deviation of annual_kWh?

- (A) 2°F (same as the input standard deviation)
- (B) $8,760 \times 1.026 = 8,988$ kWh (missing the sigma_T multiplier)
- (C) $2 \times 8,760 \times 1.026 \approx 17,975$ kWh
- (D) $166,107 / (2 \times 1.026) = 80,936$ kWh

Q5. A project manager wants to set a budget so there is a 90% probability that the actual annual energy cost will be at or below the budget. Which percentile of the Monte Carlo cost distribution should be used?

- (A) 10th percentile (P10)
- (B) 50th percentile (median)
- (C) 90th percentile (P90)
- (D) 95th percentile (P95)

Q6. A variance decomposition of annual energy cost shows that outdoor temperature uncertainty contributes 63% of total cost variance and electricity rate uncertainty contributes 36%. An engineer has funds for one improvement. Which investment reduces cost uncertainty more?

- (A) Improve occupancy scheduling software (reduces schedule uncertainty)
- (B) Negotiate a fixed-rate electricity contract (reduces rate uncertainty, contributes 36%)
- (C) Upgrade outdoor temperature sensors to reduce measurement uncertainty (contributes 63%)
- (D) Both B and C reduce cost variance by the same amount

Q7. A Monte Carlo simulation produces 10,000 annual cost values. The result shows $P(\text{annual_cost} > \$24,000) = 0.07$. Which interpretation is correct?

- (A) The maximum possible annual cost is \$24,000
- (B) The mean annual cost is $\$24,000 \times 0.07 = \$1,680$
- (C) In approximately 7% of simulated years, the annual cost would exceed \$24,000 given the assumed input uncertainties
- (D) There is a 93% probability that all future years will cost exactly \$24,000

Q8. A Monte Carlo simulation produces an 80% confidence interval (P10 to P90) of \$7,000 for annual energy cost. A second scenario (with improved temperature sensors) produces an 80% CI of \$4,000. What does the narrower interval in the second scenario indicate?

- (A) The mean cost is lower in the second scenario

- (B) The model is less accurate in the second scenario
 - (C) Reducing temperature sensor uncertainty meaningfully reduces the spread (uncertainty) in predicted annual cost
 - (D) The 80% CI width is unrelated to input uncertainty
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Q9. A student runs a Monte Carlo simulation with $N = 100$ samples and another with $N = 10,000$ samples, both with seed = 42. Which statement is true?

- (A) $N = 100$ gives more stable percentile estimates because fewer outliers distort the tails
 - (B) $N = 10,000$ gives more stable percentile estimates because tail percentiles (P5, P95) require many samples to estimate reliably; with $N = 100$ only 5 samples fall in each tail
 - (C) Both give identical results because the same seed is used
 - (D) N does not affect percentile estimates for normally distributed outputs
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Q10. OAT sensitivity analysis (Week 12) found that a $\pm 5^\circ\text{F}$ temperature bias causes $\pm 44,936$ kWh/yr and that ± 260 occupied hours causes $\pm 4,642$ kWh/yr. In the Week 13 Monte Carlo with $\text{delta_T} \sim \text{Normal}(0, 2^\circ\text{F})$ and $\text{delta_occ} \sim \text{Normal}(0, 100 \text{ hrs})$, which source of uncertainty is NOT captured by OAT but is important in the full Monte Carlo?

- (A) The nonlinearity of the cooling model
- (B) The effect of correlated regression coefficients on predictions
- (C) Electricity rate uncertainty — rate was not a regression input and so could not appear in the OAT tornado diagram, but it multiplies total kWh and contributes significantly to cost variance
- (D) The uncertainty in the building's square footage